

2019
Higher School Certificate
Trial Examination

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided
- For questions in Section II, show all relevant working in questions involving calculations
- Write your student number and/or name at the top of every page

Total marks – 100

Section I (Pages 2–13)

20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Section II (Pages 14–34)

80 marks

- Attempt Questions 21–37
- Allow about 2 hours and 25 minutes for this section

This paper MUST NOT be removed from the examination room

STUDENT NUMBER/NAME:.....

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Section I

20 marks

Attempt Questions 1–20

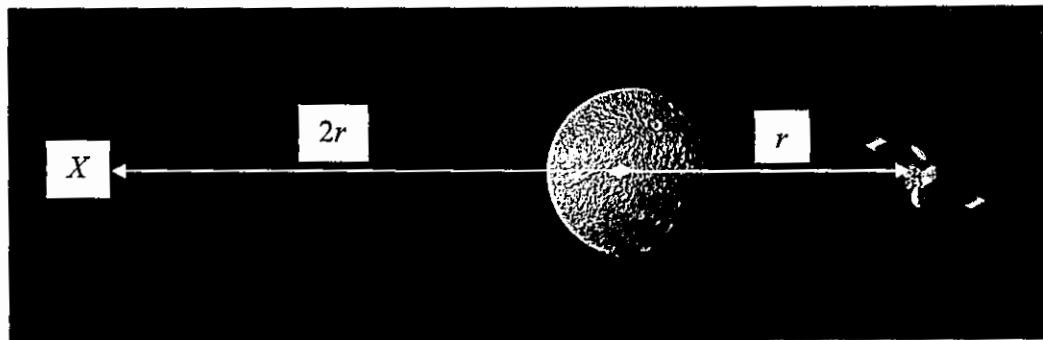
Allow about 35 minutes for this part

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
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	A	B	C	D
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- 1 Which identifies the characteristics of the motion of projectiles?
- (A) Their vertical velocity is greater than their horizontal velocity.
 - (B) Both their net velocity and acceleration are constant.
 - (C) They have constant horizontal motion and uniformly accelerated vertical motion.
 - (D) They have constant vertical motion and uniformly accelerated horizontal motion.
- 2 The diagram (not to scale) shows a satellite orbiting a planet. The orbital radius of the satellite is r . Another orbital position, X , is shown on the diagram a distance of $2r$ from the planet's centre.



How would the gravitational potential energy of the satellite change if the satellite was moved into the orbit indicated by X ?

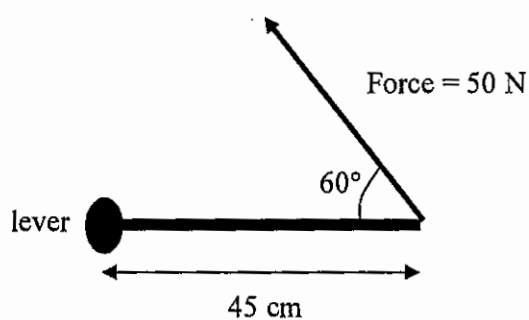
- (A) It would increase by a factor of 3.
 - (B) It would increase by a factor of 2.
 - (C) It would decrease by a factor of 3.
 - (D) It would decrease by a factor of 2.
- 3 A 250 g mass is rotating at 100 revolutions per minute on a 2.0 m long string.
- How would the angular velocity of this mass change if it continued to rotate with the same frequency, but the string was let out until it was 3.0 m long?
- (A) The angular velocity would be 1.5 times greater.
 - (B) The angular velocity would decrease by one third.
 - (C) The angular velocity would decrease by two thirds.
 - (D) The angular velocity would not change.

- 4 Consider the information about three planets shown in the table.

<i>Planet</i>	<i>Mass (kg)</i>	<i>Diameter (km)</i>	<i>Distance from Sun ($\times 10^6$ km)</i>
<i>X</i>	3.3×10^{23}	4879	57.9
<i>Y</i>	4.87×10^{24}	12 104	108.2
<i>Z</i>	6.42×10^{23}	6792	227.9

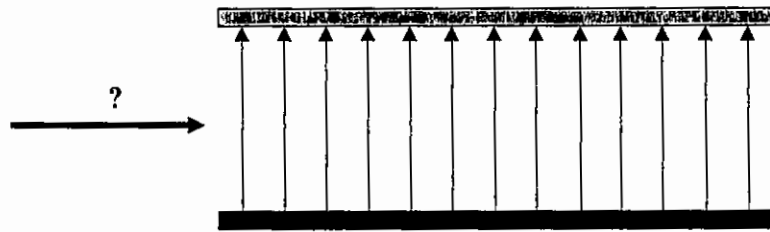
Based on this information, which choice best predicts the relative size of the escape velocity for each planet?

- (A) $X > Y > Z$
 (B) $Y > Z > X$
 (C) $Y > X > Z$
 (D) $Z > Y > X$
- 5 What is the magnitude of the net torque on the lever?



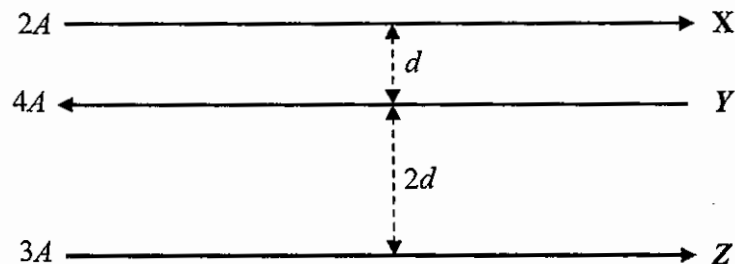
- (A) 11.25 Nm
 (B) 19.5 Nm
 (C) 1125 Nm
 (D) 1948.5 Nm

- 6 Consider the electric field between two parallel plates.



Which particle, fired horizontally from the left and moving into the field, would experience the greatest electric force downwards?

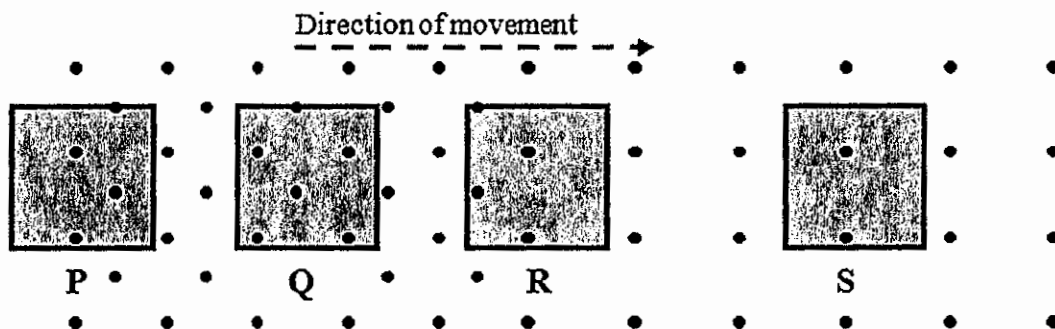
- (A) An electron
 - (B) A proton
 - (C) An alpha particle
 - (D) A magnesium ion, Mg^{2+}
- 7 Three current-carrying wires are set up as shown. The force on wire Y due to the other two wires in this situation is F .



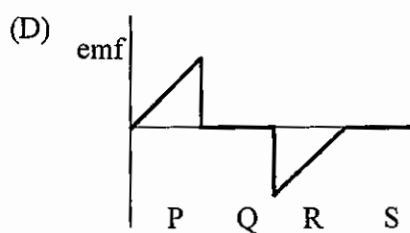
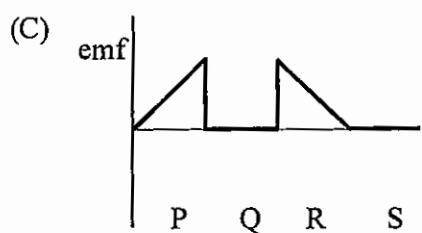
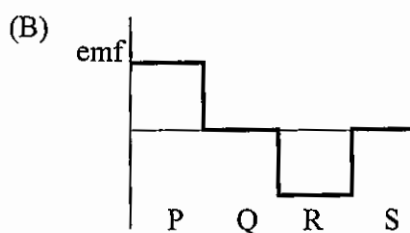
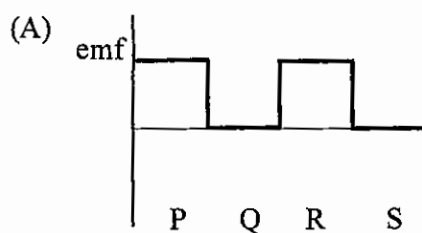
What will happen to the force on Y if the currents in X and Z are each doubled but the distances stay the same?

- (A) It will halve to $F/2$.
- (B) It will be the same value but in the opposite direction.
- (C) It will double to $2F$.
- (D) It will quadruple to $4F$.

- 8 Consider the square copper plate moving at constant speed through the magnetic field.



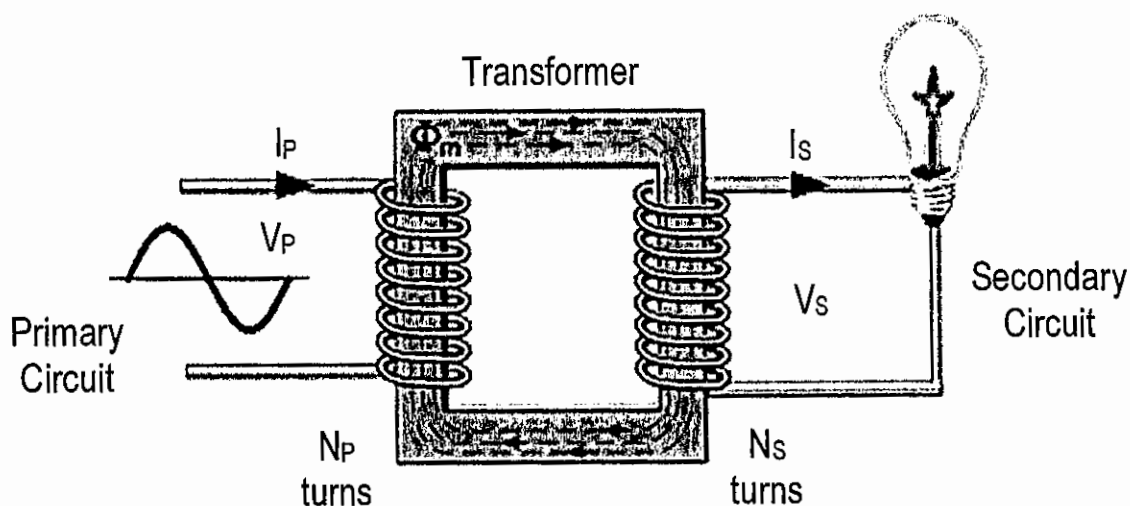
Which graph best shows the emf induced in the plate due to its movement through the field?



- 9 Which of the following is NOT an application of the motor effect?

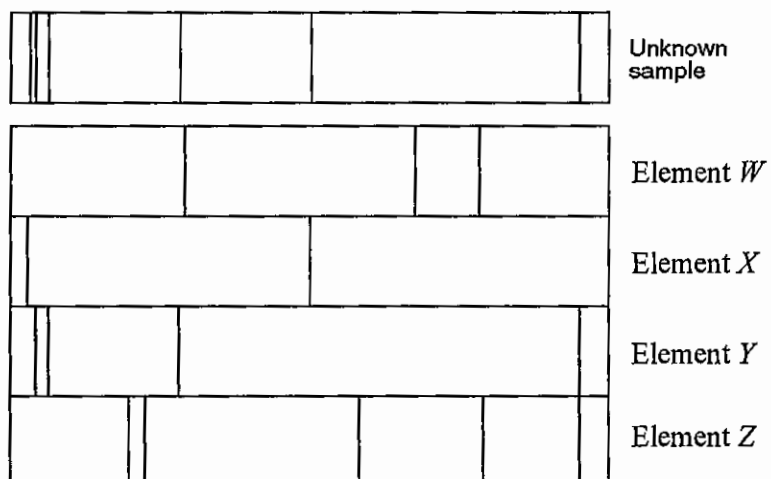
- (A) Back emf in motors
- (B) Magnetic braking systems
- (C) Torque on a coil in a motor
- (D) Transformers

- 10 Consider the following ideal transformer with 1000 turns in the primary and 1125 turns in the secondary.



- If the input voltage is 240 V and current 5 A, what is the voltage across the globe and the current through it?
- (A) 213.3 V, 5.6 A
(B) 213.3 V, 4.4 A
(C) 270 V, 5.6 A
(D) 270 V, 4.4 A
- 11 What essential feature of the structure of matter allows us to use the atomic spectra of elements to identify them?
- (A) Electron orbits have quantised energy values.
(B) The number of electrons is different for each element.
(C) The number of electrons that can fit into any orbit is given by $2n^2$.
(D) Electrons can absorb or release energy when they change levels.

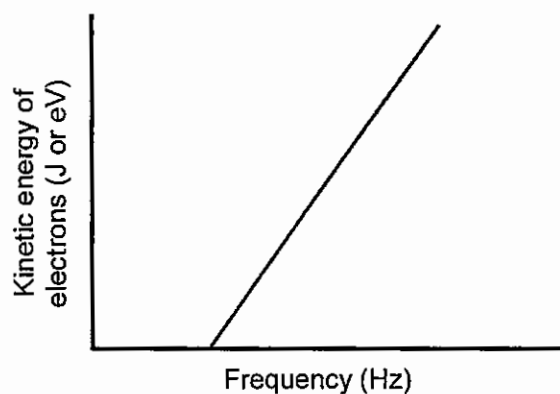
12 Consider the simple line spectra.



Which of the elements *W*, *X*, *Y* and *Z* are contained in the unknown sample?

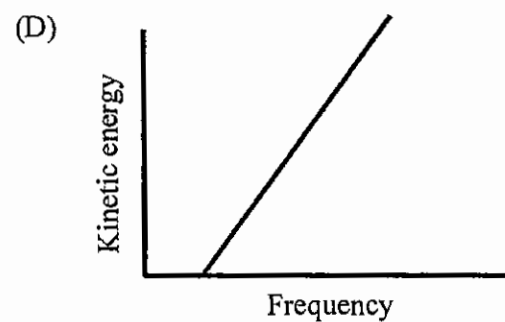
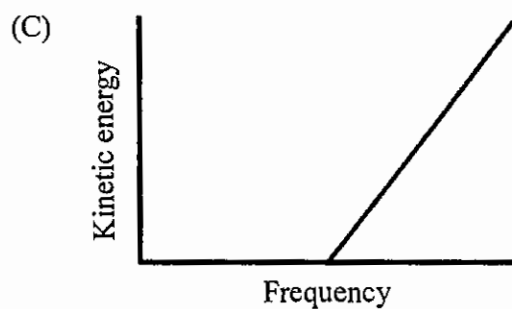
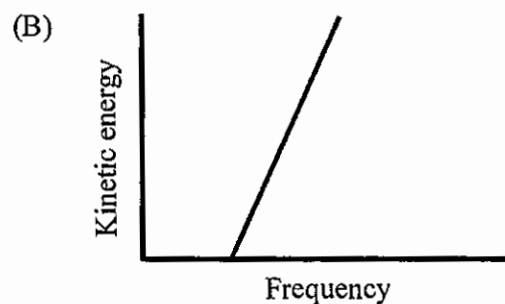
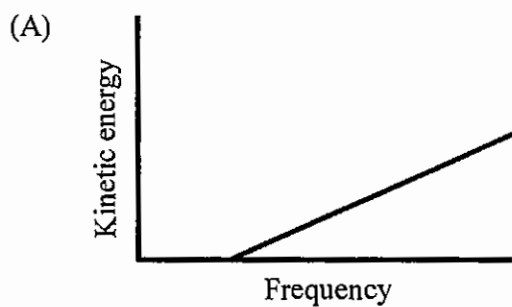
- (A) Only element *X*
- (B) Only elements *X* and *Y*
- (C) Only elements *W* and *Z*
- (D) Only elements *X*, *Y* and *Z*

- 13 The graph shows the relationship between the kinetic energy of photoelectrons emitted from a surface and the frequency of the incident light.

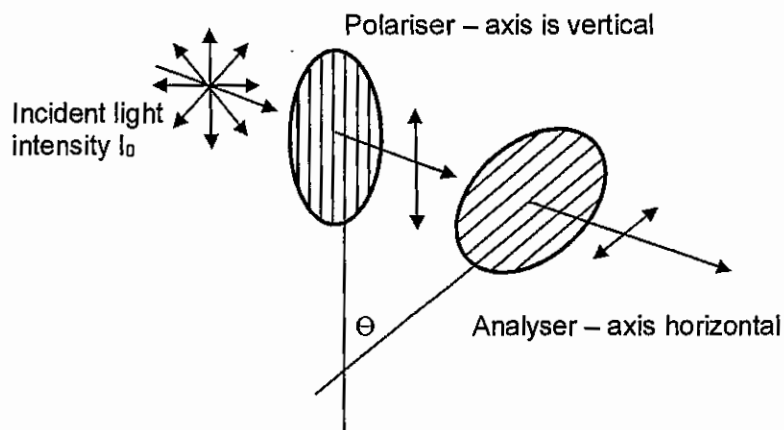


Which of the following graphs correctly shows the same relationship for a different photo-emitter that has a threshold frequency less than the graph above?

(The graphs below have identical scales)

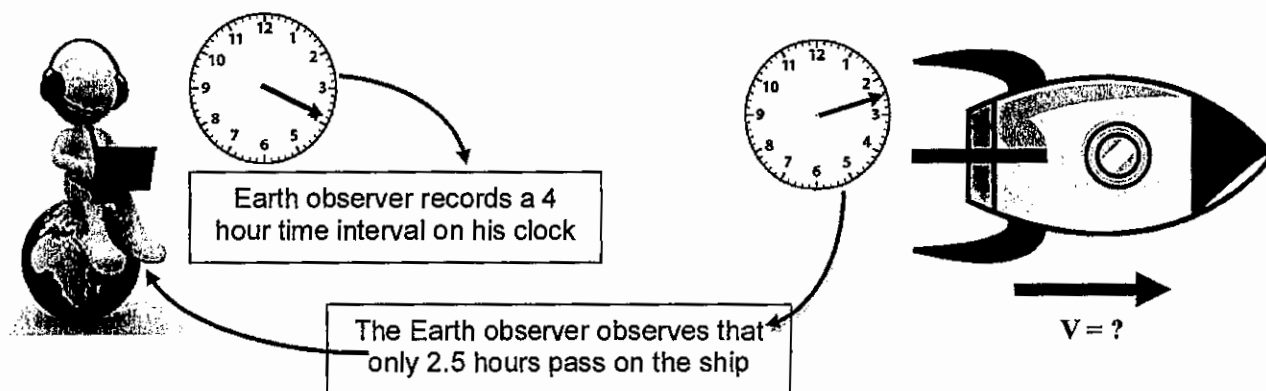


- 14 The diagram shows a polarising filter which has its polarising axis vertical and an analyser which has its polarising axis at an angle (Θ) to the first.



What value for angle Θ will allow only 40% of the incident light intensity to pass through the analyser?

- (A) 20°
 (B) 26.5°
 (C) 31.9°
 (D) 50.7°
- 15 The diagram shows an observer on Earth watching a space ship travel away from him.

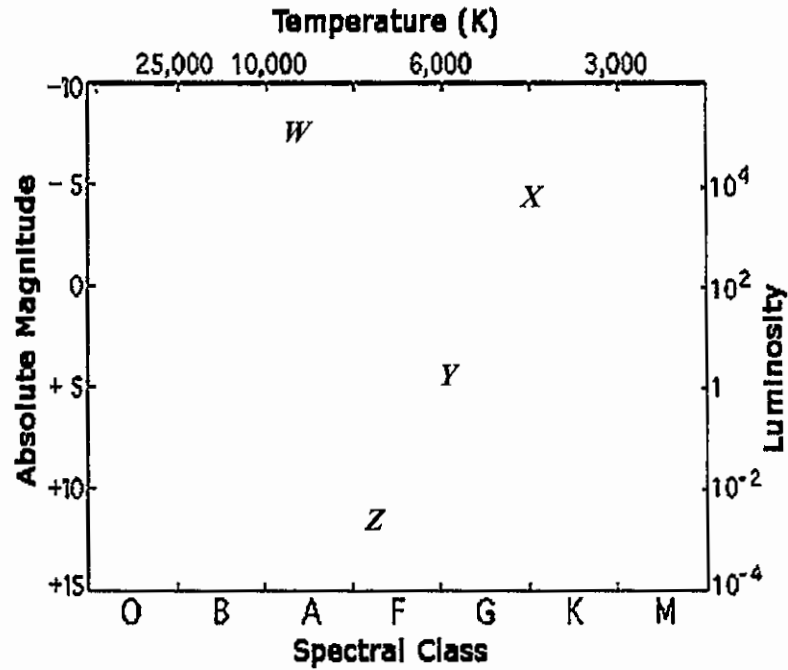


According to this information, how fast is the ship travelling?

- (A) $0.61 c$
 (B) $0.625 c$
 (C) $0.78 c$
 (D) $1.2 c$

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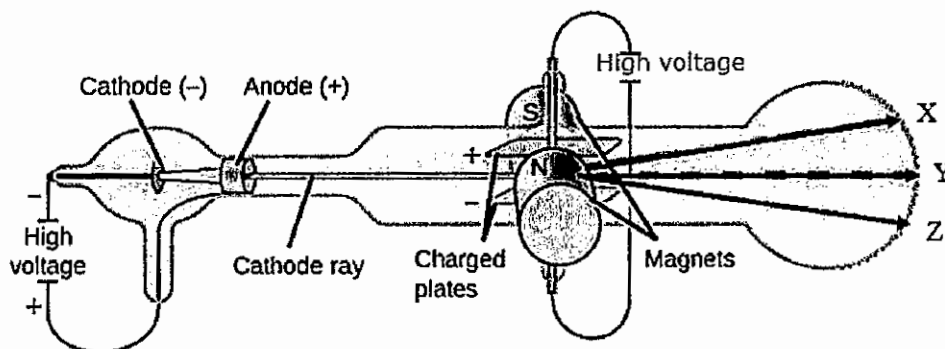
- 16 Consider the Hertzsprung-Russell diagram of Four stars, *W*, *X*, *Y* and *Z*.



Which statement about these stars is correct?

- (A) *W* is the most massive star.
- (B) *X* is a main sequence star.
- (C) *Y* will be bluish-yellow in colour.
- (D) *Z* is a red dwarf.

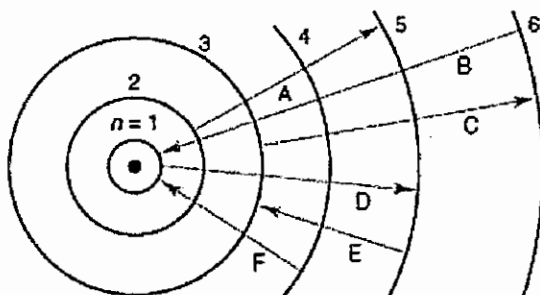
- 17 The diagram shows a discharge tube similar to those used by Thomson in his experiments with cathode rays. Three impact positions of the cathode rays, X, Y and Z are labelled on the screen.



Which choice correctly identifies these impact points?

	<i>Point X</i>	<i>Point Y</i>	<i>Point Z</i>
(A)	Deflection by electric field only	No net deflection	Deflection by magnetic field only
(B)	Deflection by magnetic field only	Deflection by electric field only	No net deflection
(C)	Deflection by electric field only	Deflection by magnetic field only	No net deflection
(D)	Deflection by magnetic field only	No net deflection	Deflection by electric field only

- 18 The diagram shows a representation of an atom.



What is this diagram showing?

- (A) Energy being released as electrons fall to lower levels on the Balmer model
 - (B) The particle nature of electrons according to De Broglie
 - (C) Quantised energy level transitions within the Bohr model of the atom
 - (D) Schrödinger's contribution to the model of the atom
- 19 Atom X undergoes alpha decay to form atom Y .
- How will the nucleus of Y be different from that of X ?
- (A) The nucleus of Y will have 2 protons more than the nucleus of X .
 - (B) The nucleus of Y will have 2 protons less but the same number of neutrons as the nucleus of X .
 - (C) The nucleus of Y will have 2 protons and 2 neutrons more as the nucleus of X .
 - (D) The nucleus of Y will have 2 protons and 2 neutrons less as the nucleus of X .
- 20 In the Standard Model of matter, which particles are the fundamental particles?

- (A) Quarks and fermions
- (B) Quarks, electrons and neutrinos
- (C) Quarks, electrons and hadrons
- (D) Fermions, baryons and hadrons

Section II**80 marks****Attempt Questions 21–37****Allow about 2 hours and 25 minutes for this part**

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 21 (3 marks)**Marks**

On planet X , much smaller than Earth, a projectile is fired at 40 m s^{-1} at an angle of 60° to the horizontal and lands 17.67 s later at the same level from which it was launched.

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Calculate the maximum height reached by the projectile above the launch point.

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Question 22 (4 marks)**Marks**

Students carried out an experiment measuring the velocity of a 100 g object in uniform circular motion for six different radius magnitudes, keeping the centripetal force constant. Their results are shown in the table below.

4

<i>Radius of turn (cm)</i>	<i>Velocity ($m s^{-1}$)</i>
10	1.48
20	2.10
30	2.60
40	2.97
50	3.32
60	3.63

Their challenge was to use all the results to determine the constant centripetal force.

List and explain the steps of the analysis required to calculate the centripetal force. (You are not required to calculate the force.)

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Question 23 (7 marks)**Marks**

- (a) Derive an equation for the orbital velocity v of a satellite orbiting a planet of mass M at a radius r .

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- (b) Show that the total energy of an orbiting satellite is half its gravitational potential energy.

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- (c) Explain why the gravitational potential energy of an orbiting satellite is negative.

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Question 24 (3 marks)

Marks

The table shows information about Earth and Jupiter.

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	<i>Earth</i>	<i>Jupiter</i>
Mass (kg)	6×10^{24}	1.9×10^{27}
Orbital radius (km)	150 000 000	778 000 000
Orbital radius (AU)	1.0	<i>X</i>
Orbital period	365 days	<i>Y</i>

Using the values above, calculate the quantities *X* and *Y* in the table.

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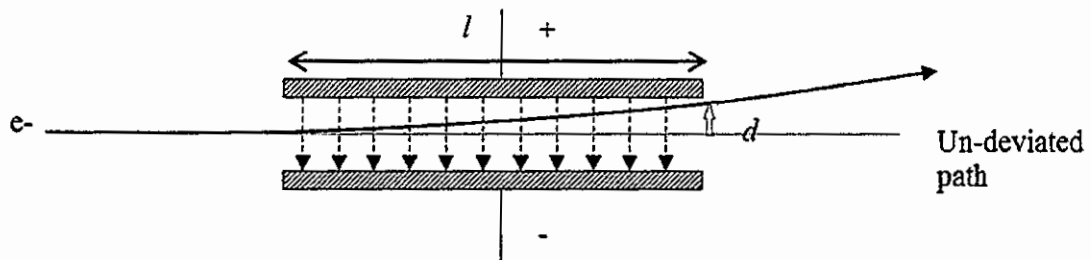
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Question 25 (4 marks)**Marks**

- (a) The diagram below shows the path of an electron, charge e , in the electric field E between parallel charged plates length l . The electron, mass m , is deviated from its path by a distance d . **1**



Derive an expression for the vertical component of the electron's velocity as it leaves the electric field.

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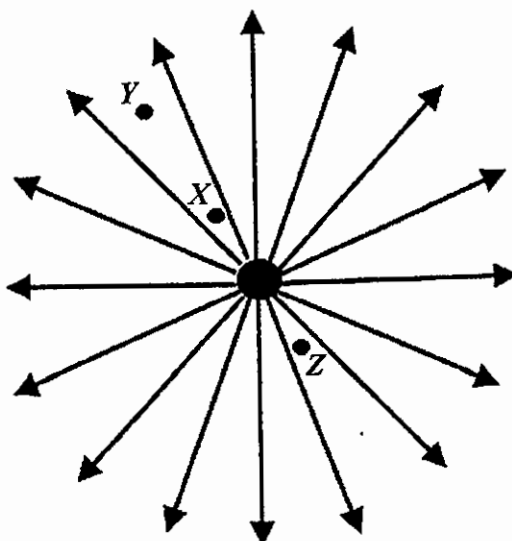
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Question 25 continues on the next page

Question 25 (continued)

Marks

- (b) Consider the following diagram which shows three points (X , Y and Z) in the electric field around a point charge.



- (i) What is the sign of the charge in the diagram? Justify your answer. 1
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- (ii) Compare the work needed to move identical negative charges from X to Y and from X to Z . Justify your answer. 2

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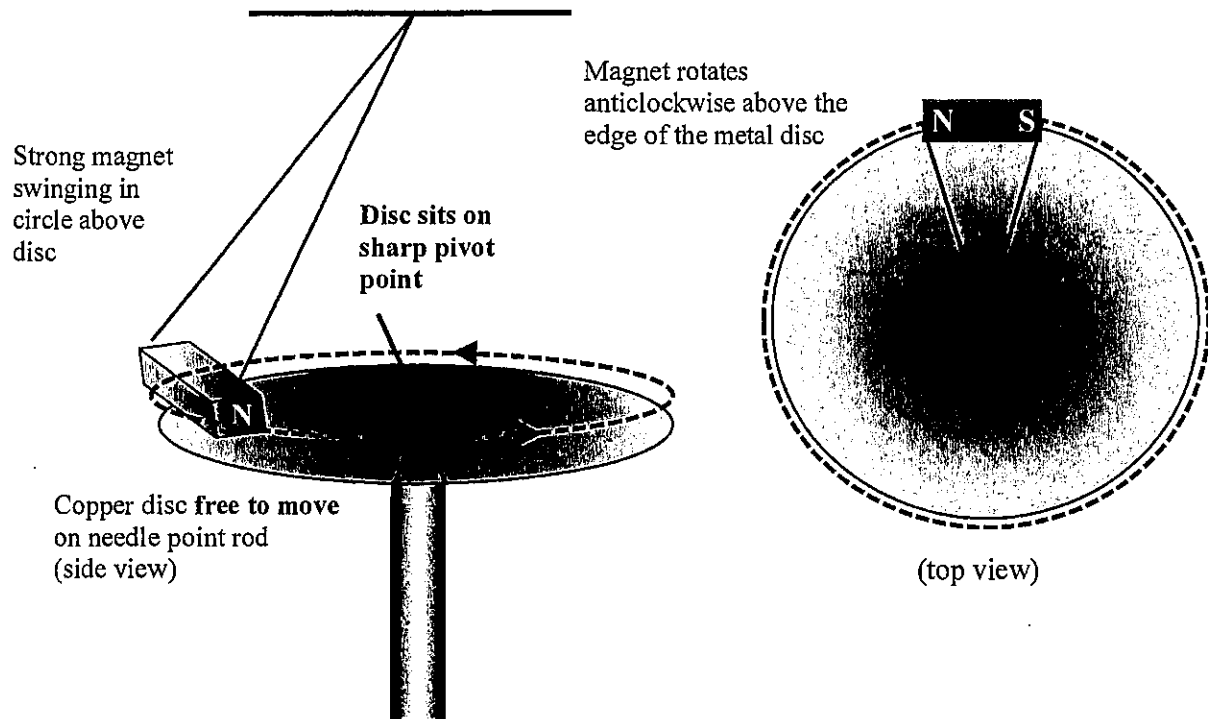
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End of Question 25

Question 26 (4 marks)**Marks**

Consider a strong magnet suspended from the ceiling and moving in a circle just slightly above a thin copper disc which is resting on a sharp pivot point, as shown in the diagram. **4**



In terms of the principles of physics involved, describe and explain what happens as the magnet rotates around above the disc.

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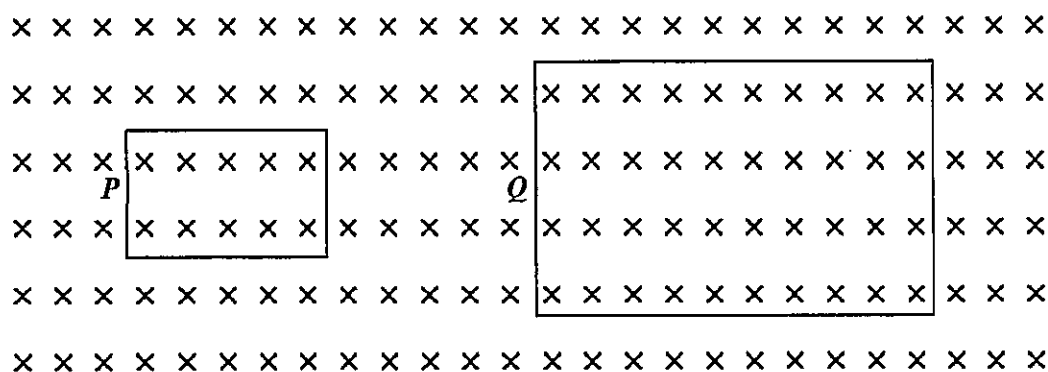
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Question 27 (3 marks)**Marks**

Two rectangular, single turn coils of wire, P and Q are inside a magnetic field as shown in the diagram.

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Compare the magnetic flux and flux density through each coil. Justify your answer.

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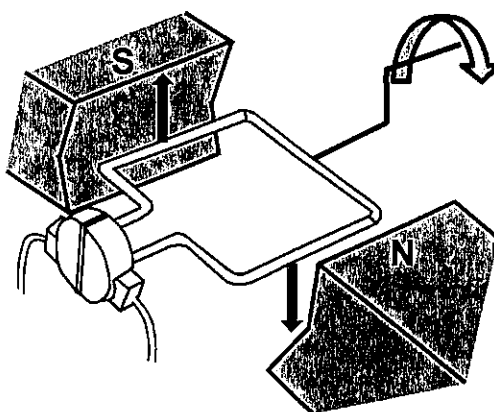
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Question 28 (4 marks)**Marks**

The diagram shows an electrical device where the coil is being rotated by an external force in the direction shown by the arrows (clockwise).



- (a) What is this device? Justify your answer.

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- (b) On the diagram, draw arrows to show the direction of current induced in the coil at the instant shown.

1

Question 28 continues on the next page

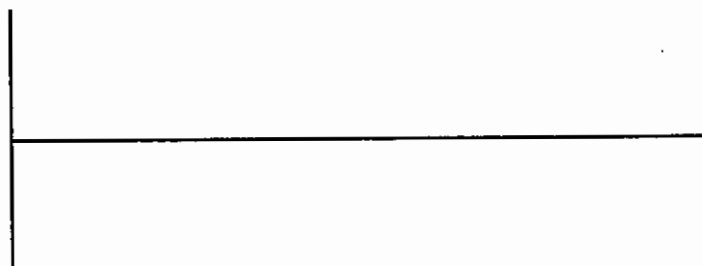
Question 28 (continued)

Marks

- (c) On the axes, sketch separate graphs to show the emf produced in the device
AND the emf delivered to the external circuit for TWO complete rotations of the
coil.

2

Emf produced in the device



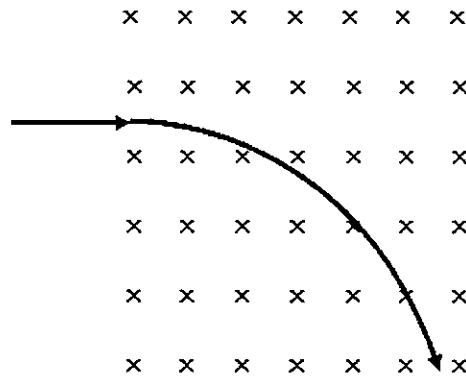
Emf delivered to external circuit



End of Question 28

Question 29 (4 marks)**Marks**

Scientists fired a beam of charged particles into a magnetic field of strength 0.8 T at $3 \times 10^5 \text{ m s}^{-1}$. They measured the radius of curvature of the beam as 0.15 m .



- (a) The scientists concluded that the beam was composed of electrons.

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Using the data given, evaluate their conclusion.

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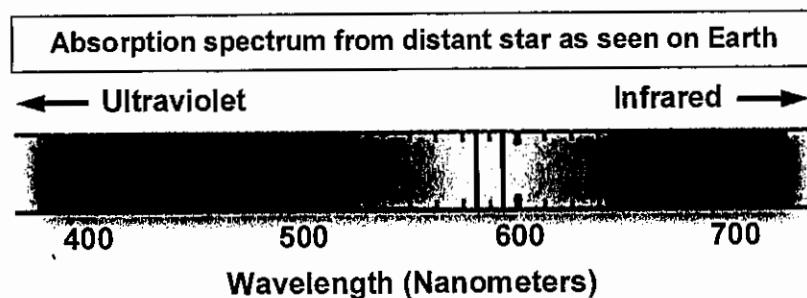
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- (b) On the diagram above, show the path the beam of charged particles would take as it left the magnetic field.

1

Question 30 (5 marks)**Marks**

The diagram shows the absorption spectrum of light received from a distant star. There are only two lines in this spectrum.



- (a) Scientists on Earth deduced that the star is moving away from Earth.

4

Explain how an absorption spectrum is formed and how scientists are able to determine the direction of movement of the star.

- (b) On the diagram below, draw the two spectral lines in a position they could be observed by scientists if they were on a planet orbiting this star.

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Question 31 (7 marks)

Marks

Compare the models of light that were proposed by Newton and Huygens and analyse the experimental evidence that supported the models at the time.

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Question 32 (3 marks)**Marks**

Muons travel through the atmosphere at a speed of $0.998c$. The mean lifetime of a stationary muon is $2.2 \mu\text{s}$.

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Show that muons formed at an altitude of 9 km can, despite their life of $2.2 \mu\text{s}$, reach the surface of the Earth.

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Question 33 (8 marks)

Marks

Evaluate Einstein's contribution to the development of special relativity and the evidence confirming or denying the postulate that the speed of light in a vacuum is an absolute constant.

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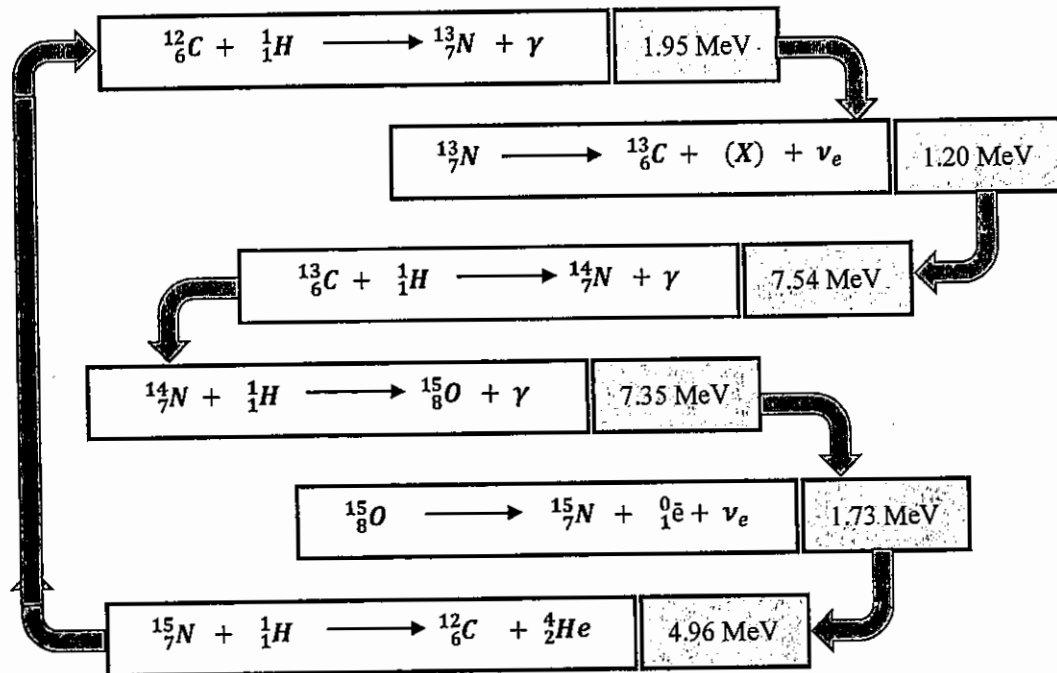
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Question 34 (5 marks)**Marks**

The diagram below shows one of the reaction sequences in the CNO cycle, which is the main energy producing sequences in stars at least 1.3 times larger than our Sun. Note that one product of the second reaction in the cycle is missing, (X).



- (a) What are the reactants and products of the CNO cycle? Justify your answer.

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- (b) Identify the missing particle, (X).

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- (c) Why is only 2% of the helium fused in the Sun using the CNO cycle?

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Question 35 (7 marks)**Marks**

- (a) Rutherford's model of the atom was based on results of experiments carried out by Geiger and Marsden.

Justify Rutherford's conclusions that were the basis of his model

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- (b) Draw a labelled diagram of the model of the atom proposed by Rutherford following these experiments and a labelled diagram of the model it replaced.

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Rutherford model

Previous model

- (c) Outline the main problem with the Rutherford model.

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End of Question 35

Question 36 (3 marks)

Marks

Outline Schödinger's model of the atom.

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Question 37 (6 marks)

Before an experiment to find the half-life of a radioactive isotope, a nuclear scientist switched on her radiation counter in the lab for five 60 second intervals and recorded the following results:

<i>Radiation recorded (counts/min⁻¹)</i>
1123
1126
1128
1127
1121

The scientist then placed a sample of radioactive isotope *X* near the counter and measured the counts per minute each second Monday for 14 weeks. Counting the first week as week 0, (the starting week), her results are shown in the table below.

<i>Time (weeks)</i>	<i>Total radiation counts recorded (counts min⁻¹)</i>
0	11 125
2	7725
4	5525
6	4125
8	3125
10	2425
12	2075
14	1625

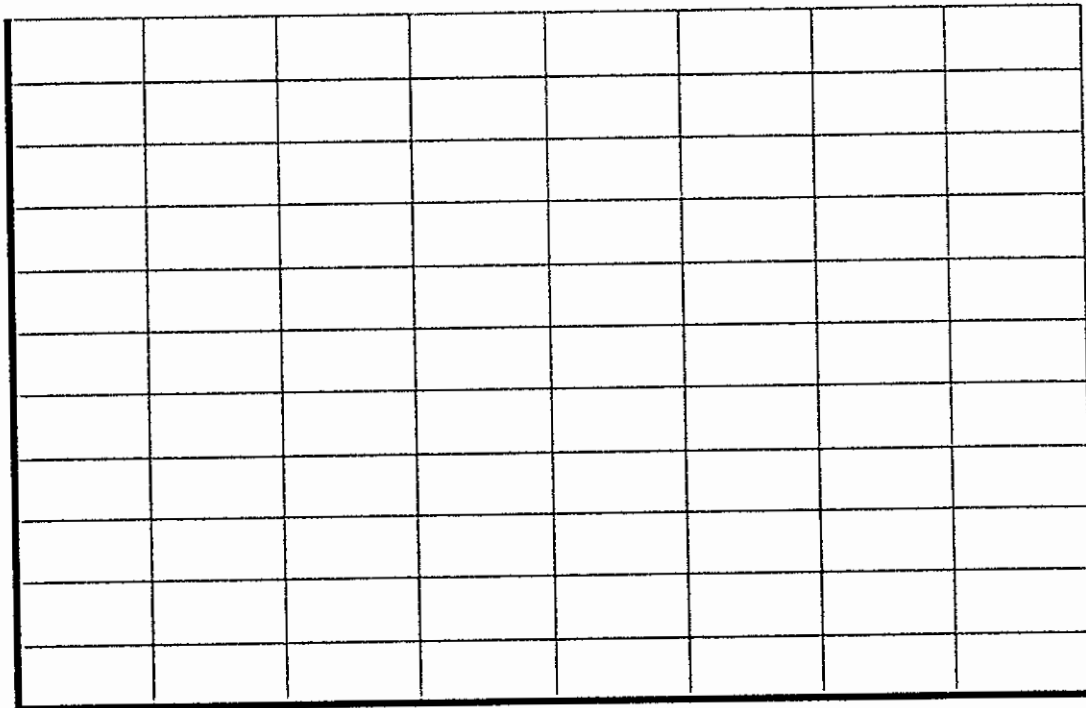
Question 37 continues on the next page

Question 37 (continued)

Marks

- (a) Plot results to show the relationship between time and the radiation of X.

4



- (b) Use your graph to determine the half-life of X.

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- (c) Identify an assumption made by the scientist.

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End of paper

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Extra writing space

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